

Problem Set 0: Scavenger hunt

Physics 230A/146A — Biological Physics

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Find values for the following biological quantities. You may use Google, Wikipedia, textbooks, peer-reviewed papers, large language models, or anything else, but you must cite your sources¹. One valuable source is: bionumbers.org. In cases where answers need to be qualified, do so. Are there ranges? Are there a few exceptions or many exceptions? Are these order-of-magnitude estimates?

1. The number of amino acids in the average human protein

The size of (in meters or other SI unit):

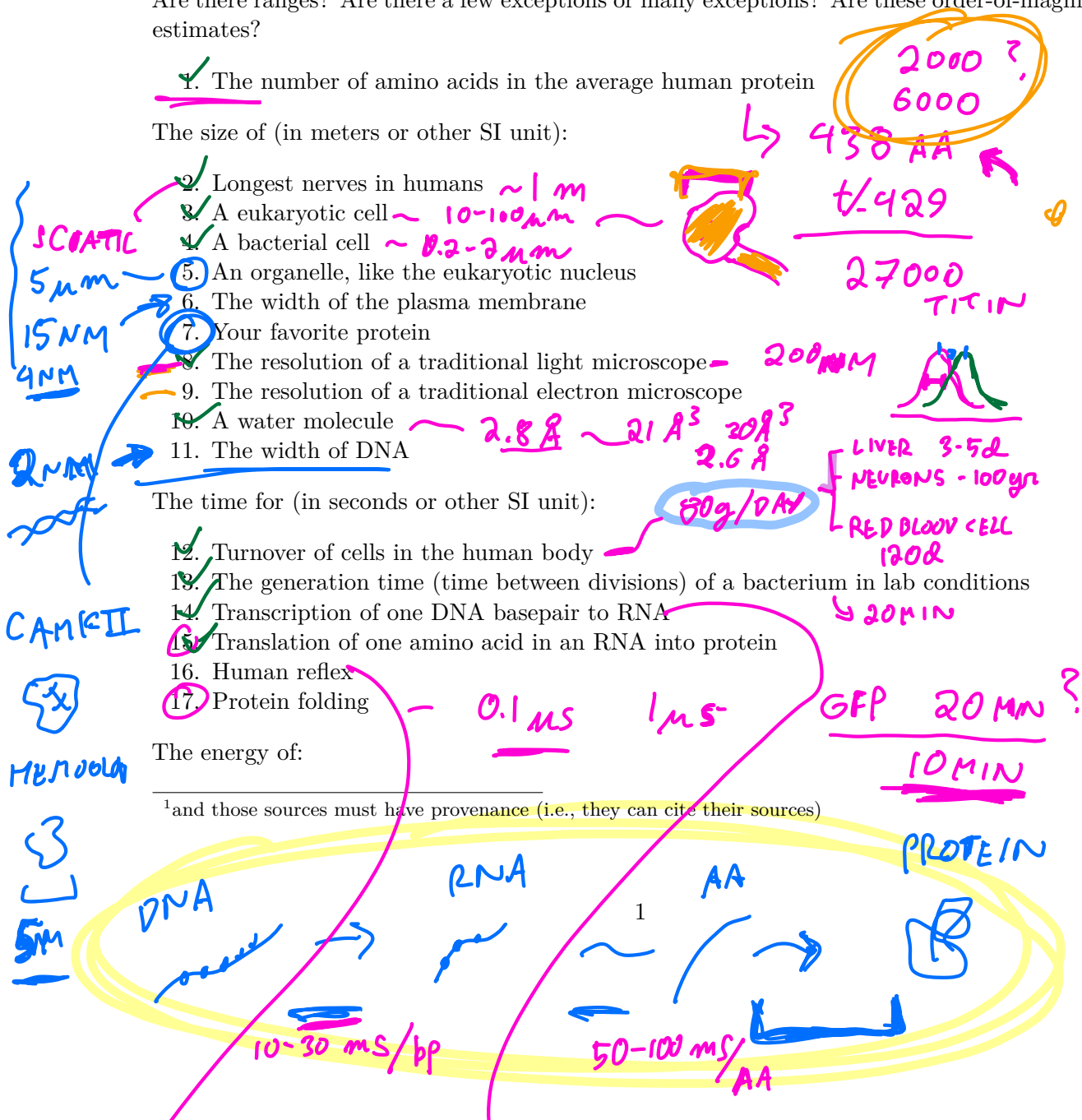
2. Longest nerves in humans $\sim 1\text{ m}$
3. A eukaryotic cell $\sim 10-100\mu\text{m}$
4. A bacterial cell $\sim 0.2-2\mu\text{m}$
5. An organelle, like the eukaryotic nucleus
6. The width of the plasma membrane
7. Your favorite protein
8. The resolution of a traditional light microscope $\sim 200\text{ nm}$
9. The resolution of a traditional electron microscope
10. A water molecule $\sim 2.8\text{ \AA}$
11. The width of DNA $\sim 2\text{ nm}$

The time for (in seconds or other SI unit):

12. Turnover of cells in the human body
13. The generation time (time between divisions) of a bacterium in lab conditions
14. Transcription of one DNA basepair to RNA
15. Translation of one amino acid in an RNA into protein
16. Human reflex
17. Protein folding

The energy of:

¹and those sources must have provenance (i.e., they can cite their sources)



$$30-45 \text{ kJ/mol} \rightarrow$$

$$5 \times 10^{-20} \text{ J}$$

18. The reaction $\text{ATP} \rightarrow \text{ADP-Pi}$

19. The thermal energy unit $k_B T$ (where k_B is Boltzmann's constant and T is the temperature from absolute zero)

20. Moving one electron against the electric field across the membrane of a cell

OCULAR 8-12 ms
TOUCH 100 ms

$$V \sim 20-200 \text{ mV}$$

$$E \sim 10^{-20} \text{ J} \sim 20-200 \text{ meV}$$

$$E \sim 9 \times 10^{-21} \text{ J}$$

FREE

$$T \sim 293-300 \text{ K}$$

